

instructions given below to set up Android Studio.

1. Once you launch *Android Studio.exe*, you can see the screenshot shown in Figure 1, which includes the installation for Android Studio, Android SDK, Android Virtual Machine and the Performance (Intel chip).
2. Choose the location to store Android Studio and Android SDK.
3. Configure the emulator setup. By default, the allocated RAM is 512MB.
4. Now it will fetch all the SDK packages into our local machine and install Android Studio.
 - Before developing any app, you need to install Android SDK packages. To download the SDK packages, follow the steps given below:
 1. Go to *Android Studio > Configure option > SDK Manager*.
 2. Some of the packages are already selected. Apart from those, choose some other required tools and packages from the following choices:
 - Android SDK tools
 - Android SDK platform-tools
 - Android SDK build-tools (highest version available)
 - SDK platform (from the latest Android version folder)
 - A system image for the emulator, such as the ARM EABI v7a system image
 3. Install the packages chosen for the selected tools and packages.
 - To run and test the developed Android App, we need an emulator. Create an emulator as follows.

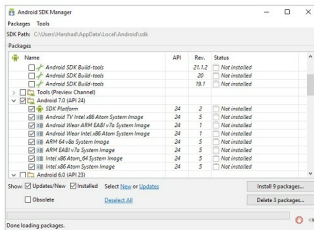


Figure 2: Android SDK manager

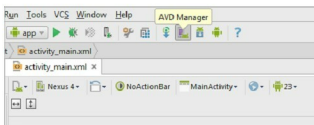


Figure 3: AVD manager



Figure 4: Running the AVD manager with configured devices

1. Click on the *AVD manager* icon in Android Studio.
2. Click on the *Create* button to create a new emulator. You can see all the available AVDs here.
3. Give the proper details to the AVD, and you will be able to see your AVD in the *AVD manager*.
4. Once you start the AVD, it will display the screen shown in Figure 4, as seen in an Android mobile.

The 'Hello World' Android app

To make the sample *Hello World* Android app in Android Studio, follow the steps given below:

1. Add the new Android project in Android Studio.
2. Give the project details and the location where you want the Android projects stored.
3. Specify the minimum SDK for the app.
4. Select the blank activity.
5. Insert the activity's name and details.

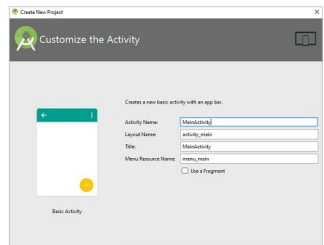


Figure 5: Activity configuration

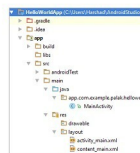


Figure 6: Project structure

Files/components structure

Given below are the important files and directories of Android projects and their functions.

1. *src*: This contains the main *java* source file, which is the primary activity class that is being executed when the app is launched.
2. *bin*: This contains the *.apk* file for our project, which stores all the packages needed while running the app.
3. *res/layout*: This is the directory that contains the app's layout files.